

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I M. Mohan/191233011 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Mohan

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M. Mohan siva reddy
192573811
CSA1714
Artificial Intelligence

1) Cybersecurity Intrusion detection system

(a) Propositional logic representation

Let:

• F = Failed login Attempts is True.

• U = Unusual IP Address

• O = odd login time

• H = High data transfer

• Rules : $F \wedge U \rightarrow B$, $F \wedge O \rightarrow C$, $U \wedge H \rightarrow D$

(b)

• Facts: F, U, H

• $F \wedge U \rightarrow$ Brute Force Attack

• $U \wedge H \rightarrow$ Data Exfiltration

• O is absent, so account compromise not derived.

(c)

• Goal : Data Exfiltration.

• Rule! $U \wedge H \rightarrow D$

• U and H are facts

• Therefore, Data Exfiltration is proved.

2) Airport landing system

a)

- Rule:

Aircraft x $\wedge$ Emergency Declared $(x) \rightarrow$ priority clearance (x)

- For flight Y : $x = \text{Flight } Y$

- Therefore, priority clearance $(\text{Flight } Y)$.

b)

- Goal: Delayed $(\text{Flight } Y)$

- Fact already contains Delayed $(\text{Flight } Y)$.

- Negate goal: $\neg$ Delayed $(\text{Flight } Y)$

- Resolution gives $\square$.

- Hence proved.

c)

- current rules cannot handle two emergency aircraft.

- Need a priority/landing-order rule.

- Example:

Higher priority $(x, y) \rightarrow$ priority clearance (x) .

Maintenance Expert system

a)

- P1: MachineOverheating $\vee$ Maintenance Needed
- P2: Maintenance Needed $\vee$ - High priority line $\vee$
ScheduleImmediateRepair $\vee$ - High priority line $\vee$
- P3: ScheduleImmediateRepair $\vee$ - High priority line $\vee$
ProductionRisk

b)

- MachineOverheating $\rightarrow$ Maintenance Needed
- Maintenance Needed + High priority line $\rightarrow$
ScheduleImmediateRepair
- Negated goal causes contradiction
- Therefore, ScheduleImmediateRepair is proved.

c)

- Remove MachineOverheating.
- Maintenance Needed cannot be derived
- ScheduleImmediateRepair cannot be proved.
- Therefore, ProductionRisk may occur.

4) Minefield Robot

a)

- vibration $[2,1] \rightarrow$ mine nearby
- smoke $[1,1] \rightarrow$ Fire nearby
- Beacon $[2,2] \rightarrow$ safezone Marker $[2,2]$

b)

- possible mine locations: $[1,1], [3,1], [2,2]$
- possible fire locations: $[2,1], [1,2]$
- safe-zone marker: $[2,2]$

c)

• path $[1,1] \rightarrow [2,1] \rightarrow [2,2]$

• Mine/fire risks are present.

• Negative facts proving safety are absent

• Therefore, path is not confirmed safe.